Fidelity Is Not Utility: A Controlled Dissociation in Learned Image Translation

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Abstract

Generative image translation—harmonization, style transfer, synthetic-to-real adaptation—is almost universally evaluated with *distributional fidelity* metrics such as the Fréchet Inception Distance (FID). We show, in a controlled setting, that improving fidelity does not imply improving downstream utility, and that for *learned* translation the two systematically trade off. Holding a downstream model *frozen*, we measure a fidelity axis (FID to the target domain) and a utility axis (the frozen model's task metric) on identical inputs. Across two domains (synthetic-to-real driving and multi-site brain-tumor MRI) and two generative families (adversarial CycleGAN and diffusion SDEdit), learned translation lowers FID while degrading the downstream task metric, whereas a non-learned baseline (histogram / color transfer) preserves the task metric without improving FID. Using SDEdit's noise-strength parameter as a continuous knob, we trace a monotonic FID–utility frontier ($N{=}1000$, three seeds): no learned setting reaches the high-fidelity, high-utility region, and none Pareto-dominates the non-learned baseline. An empty-prompt ablation shows the utility loss persists even when fidelity is *not* improved, isolating the effect to the learned generative prior rather than any conditioning signal. A structure-preserving baseline (ControlNet conditioned on edges) does not rescue utility either: it lands on the learned frontier, confirming that generic structural conditioning is insufficient—only injecting the task labels themselves recovers utility, which concedes our thesis. Our contribution is not a new method but a *measurement*: a general, architecture-agnostic dissociation between what fidelity metrics reward and what downstream tasks need.

1 Introduction

Distribution-matching translation is trained and evaluated to make outputs *look like* a target domain. The implicit promise is that looking more like the target makes outputs more *useful*—for example, that harmonizing site B to site A helps a model trained on A. We test that promise directly and find it false for learned translation.

Our contributions are:

- A **controlled protocol** for separating fidelity from utility: a frozen downstream evaluator plus a matched non-learned control, measuring FID and the task metric on identical inputs (Section 3).
- Evidence that the fidelity–utility dissociation is a property of the **learned distribution-matching objective**, not a single architecture—it reproduces under both a GAN (CycleGAN) and a diffusion model (SDEdit).
- A **continuous characterization** via SDEdit strength: a monotonic FID-vs-mIoU frontier with tight seed variance, an empty-prompt ablation that decouples utility loss from fidelity gain, and a structure-preserving (ControlNet) baseline that fails to escape the frontier.
- **Cross-domain** generality: the same dissociation in driving scenes and in clinical brain-tumor MRI.

2 Related Work

Three separately-established observations converge in our claim; we unify them.

Fidelity metrics mis-rank generators for downstream tasks. Konz et al. [2024] show no perceptual metric reliably tracks segmentation quality in medical translation; Dutt et al. [2025] show low FID does not predict clinical utility; and Kynkäänniemi et al. [2023] show FID is gameable by aligning ImageNet-class statistics. These are *metric-evaluation* studies; we add a learned/non-learned control and a second generative family and domain.

Distribution-matching translation alters task content. Cohen et al. [2018] show CycleGAN-style matching can hallucinate or remove tumors to fit the target distribution, and semantic-consistency losses [Hoffman et al., 2018, Peng et al., 2023] exist precisely to patch this. We show that *without* such non-distributional anchoring the pure learned objective degrades utility, reframing those losses as evidence for our thesis.

Non-learned alignment rivals learned translation. Yang and Soatto [2020] reach strong segmentation adaptation with a non-learned spectral swap. We use histogram/color transfer as the analogous control and, unlike FDA, measure the fidelity axis explicitly to expose the dissociation. Richter et al. [2022] likewise treat a segmentation-stability metric as a guardrail that realism must not break.

Positioning. No prior work runs the clean 2×2 $\{$learned, non-learned$\}$ $\times$ $\{$fidelity$\downarrow$?, utility$\downarrow$?$\}$ as its thesis, ties the failure to the objective across both a GAN *and* a diffusion model, and does so across driving and medicine. This is a measurement/positioning contribution, not a new architecture.

3 Setup: Separating Fidelity from Utility

Given source images x and a translation T, we evaluate $T(x)$ on two axes. **Fidelity** is $\mathrm{FID}(T(x), \text{target})$ via clean-fid [Parmar et al., 2022]; lower is closer to the target. **Utility** is a *frozen* downstream model f (never retrained) applied to $T(x)$ and scored against ground truth; because f is fixed, any change in its score is attributable purely to T. We compare four families of T: (i) identity (raw source, reference); (ii) non-learned (histogram / color transfer); (iii) learned-GAN (CycleGAN, Zhu et al., 2017); and (iv) learned-diffusion (SDEdit, Meng et al., 2022, over a sweep of noise strengths, plus ControlNet-Canny, Zhang et al., 2023). FID is compared only at equal sample size, which it is known to bias.

4 Experiments

4.1 Driving: GTA5 $\rightarrow$ Cityscapes

The source is GTA5 [Richter et al., 2016] and the target is Cityscapes [Cordts et al., 2016]. The frozen evaluator is a Cityscapes-finetuned SegFormer-b4 [Xie et al., 2021] (19 classes); utility is mIoU. We use $N{=}1000$ GTA5 images per condition, with the Cityscapes validation set (500 images) as the fixed FID reference, and identical evaluator and clean-fid (GPU) for all conditions. SDEdit runs on Stable Diffusion [Rombach et al., 2022] and rows report mean $\pm$ std over three seeds $\{42, 43, 44\}$. Results are in Table 1 and Figure 1.

Findings. Figure 2 shows representative translations. (a) Learned translation lowers FID relative to raw while collapsing mIoU by 15–72%; CycleGAN more than halves FID (137$\rightarrow$65) at mIoU 0.167. (b) The non-learned baseline does not improve FID yet preserves mIoU (-1.8%). (c) SDEdit strength traces a **monotonic frontier** with tight seed variance ($\sigma(\mathrm{mIoU}) \leq 0.005$): as strength rises, FID falls (145$\rightarrow$88) and mIoU falls with it (0.323$\rightarrow$0.106); no learned setting reaches the ideal corner, occupied only by raw/non-learned. (d) **Empty-prompt ablation.** With no text prompt, FID is *worse* than raw ($154.2 > 137.2$) yet mIoU still collapses (0.170)—the utility loss is the diffusion prior itself, not text guidance, and occurs even absent any fidelity gain. (e) **Structure-preserving baseline.** Conditioning on the source image's Canny edges (ControlNet-Canny) does *not* rescue utility: it lands on the learned frontier (FID 133.2, mIoU $0.221{\pm}0.002$), Pareto-dominated by SDEdit at equal FID (SDEdit 0.30: FID 133.1, mIoU 0.289). Edge structure is not the task-relevant signal; only conditioning on the segmentation *map* [Peng et al., 2023] recovers utility, and only by injecting the labels—the non-distributional constraint our thesis requires. A local $N{=}200$ run reproduces the same ordering on an independent machine.

Table 1: GTA5→Cityscapes, N=1000, frozen SegFormer-b4 + clean-fid. SDEdit rows are mean ± std over three seeds. Learned translation lowers FID while collapsing mIoU; the non-learned baseline preserves mIoU without improving FID; no learned setting reaches the low-FID/high-mIoU corner.

condition	family	FID ↓	mIoU ↑	ΔmIoU
raw GTA5	baseline	137.2	0.380	—
color match	non-learned	165.9	0.363	−0.018
SDEdit 0.20	diffusion	145.1±1.8	0.323±0.003	−0.057
SDEdit 0.30	diffusion	133.1±2.1	0.289±0.002	−0.092
SDEdit 0.40	diffusion	116.4±1.1	0.237±0.003	−0.143
SDEdit 0.50	diffusion	100.3±1.4	0.194±0.005	−0.187
SDEdit 0.60	diffusion	90.8±2.0	0.150±0.001	−0.231
SDEdit 0.70	diffusion	88.0±0.4	0.106±0.001	−0.274
CycleGAN	learned-GAN	65.5	0.167	−0.214
SDEdit 0.50, empty prompt	diffusion (abl.)	154.2±1.9	0.170±0.003	−0.211
ControlNet-Canny	diffusion + edge	133.2±0.3	0.221±0.002	−0.159

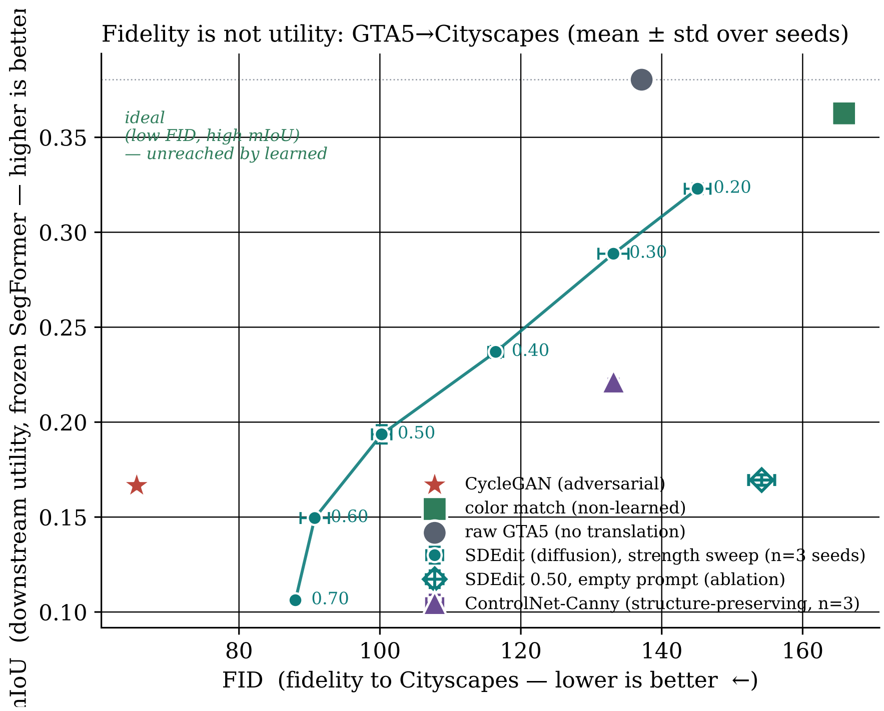

Figure 1: The fidelity–utility plane for GTA5→Cityscapes (mean ± std over seeds). The SDEdit strength sweep traces a monotonic frontier: as strength rises, FID falls and mIoU falls with it. Every learned point (CycleGAN, all SDEdit strengths, the empty-prompt ablation, and edge-conditioned ControlNet) sits well below raw and the non-learned baseline on utility; the ideal low-FID/high-mIoU corner is empty.

4.2 Medical: Multi-Site Brain-Tumor MRI

The frozen evaluator is an nnU-Net [Isensee et al., 2021] trained on BraTS [Menze et al., 2015] and frozen; utility is tumor Dice on the external UPenn-GBM site [Bakas et al., 2022] (n=103). The learned harmonizer is SA-CycleGAN-2.5D, a self-attention CycleGAN over 2.5D multi-modal MRI. Table 2 reports the result.

Findings. Learned harmonization collapses external-site Dice (0.682→0.064) while the non-learned baseline preserves it (0.669). Content/segmentation-consistency losses reduce but do not eliminate the loss ($0.205 \ll 0.682$), consistent with the driving result that the degradation is intrinsic to the learned objective. A second external site (RHUH-GBM, Cepeda et al., 2023, n=40) confirms the direction (0.718→0.599, −16.6%). We note two evaluation

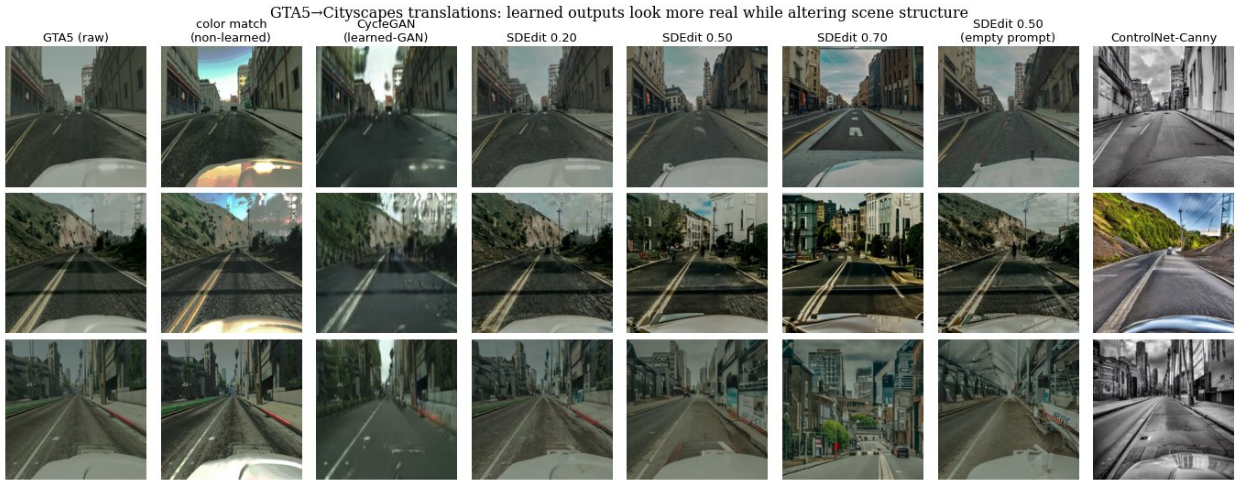

Figure 2: Qualitative GTA5→Cityscapes translations (three scenes, seed 42). Left to right: raw source, non-learned color match, CycleGAN, SDEdit at increasing strength, the empty-prompt SDEdit, and edge-conditioned ControlNet. As the learned translations become more Cityscapes-like they visibly hallucinate and alter scene structure—buildings, lane markings, added objects—that the frozen segmenter depends on, while the non-learned baseline leaves structure intact.

Table 2: External UPenn-GBM tumor Dice under a frozen BraTS-trained nnU-Net. Learned harmonization collapses Dice even though it makes images more source-like; the non-learned baseline preserves it.

input to frozen nnU-Net	family	Dice (fg. mean)
BraTS test (in-domain ref.)	—	0.771
UPenn raw (external)	baseline	0.682
UPenn + histogram match	non-learned	0.669
UPenn + learned harmonization	learned-GAN	0.064
UPenn + symmetric seg-consistency	learned-GAN (+ content)	0.205

protocols exist: the headline numbers use the frozen independent nnU-Net; a separate method-development evaluator ranks the symmetric loss as the best-preserving *learned* harmonizer, which belongs to a companion method paper.

5 Analysis

The learned objective optimizes appearance under a distribution/adversarial or denoising target with no term guaranteeing task-relevant signal survives. SDEdit strength is precisely the amount of learned prior injected, and utility falls monotonically with it. The empty-prompt result decouples utility loss from fidelity gain: passing through the prior damages task content even when the output is *less* target-like. Non-learned transforms, lacking a learned prior, cannot introduce this failure; edge-conditioned ControlNet, injecting structure but not semantics, does not avoid it.

6 Limitations

FID is sample-size biased; all conditions use the same N (1000 translated vs the 500-image Cityscapes validation set as the fixed reference), and SDEdit points report mean $\pm$ std over three seeds ($\sigma(\text{mIoU}) \leq 0.005$, $\sigma(\text{FID}) \leq 2.1$), so the ordering is robust to both sample size and seed. The structure-preserving baseline uses edge (Canny) conditioning; a segmentation-map-conditioned variant would recover utility only by injecting the labels, which concedes the thesis rather than refuting it.

7 Conclusion

Fidelity is not utility. Rewarding distributional realism can actively harm the downstream task for learned translation, across architectures and domains. We recommend reporting a frozen-evaluator utility axis alongside any fidelity metric, and treating non-learned baselines as first-class controls.

Reproducibility Statement

All conditions regenerate from a single command; seeds are fixed and logged; frozen evaluators are referenced by checkpoint id (SegFormer-b4, nnU-Net); FID uses clean-fid at a fixed N. Code, configs, and the $N{=}1000$ result set are released with the paper.

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